

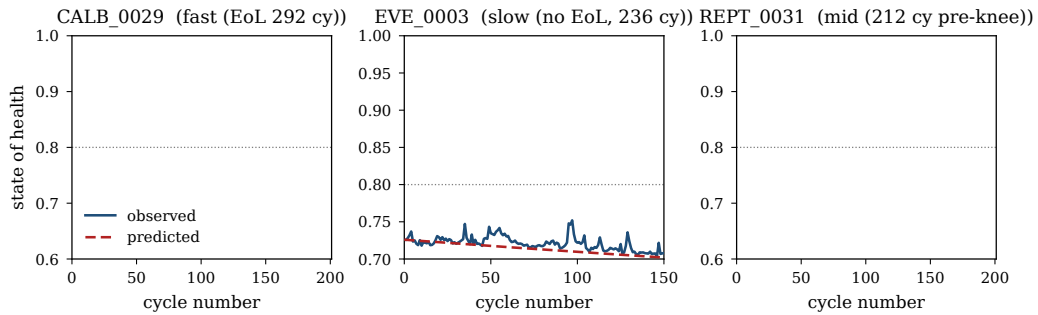
# One-hour characterisation to full state-of-health trajectory: a supplier-agnostic PyBaMM-trained neural operator for second-life LFP RUL

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Second-life reuse of used lithium iron phosphate (LFP) cells is bottlenecked by characterisation cost: reliable remaining-useful-life (RUL) estimation today requires months of accelerated cycling per cell, and existing empirical models rarely transfer between supplier chemistries. We present a supplier-agnostic neural operator that maps a single one-hour characterisation snapshot (OCV, HPPC, GITT and a partial RPT) to the full state-of-health (SoH) trajectory to end-of-life.

The operator is trained on a PyBaMM [1] simulation corpus of 490 degradation trajectories, generated by perturbing a five-parameter degradation model around seven cells drawn from three commercial LFP suppliers (CALB, EVE, REPT). Per-cell electrochemistry is identified from the characterisation snapshot; the operator itself is a  $\theta$ -conditioned Neural ODE [2] that takes the identified parameters as input and returns SoH vs cycle to the end-of-life threshold. A censored-likelihood loss makes it possible to use horizon-truncated simulations (about half of the corpus) as training signal.

We validate the trained operator on three held-out cells — one per supplier — with no per-cell degradation fit, using only a supplier-typical parameter prior. In-window SoH errors on the actual (nameplate-normalised) scale are 0.53, 1.10 and 0.97 percentage points for CALB\_0029, EVE\_0003 and REPT\_0031 respectively (Figure 1), well below the 3.0 pp acceptance threshold we set from first principles. Crucially, the standard deviation of per-cell error across the three suppliers is only 0.24 pp: for a fixed operator, prediction accuracy is essentially chemistry-independent across CALB, EVE and REPT LFP variants.



**Figure 1:** Actual SoH (discharge capacity / nameplate) on three held-out cells. Blue: observed cycling. Red dashed: operator prediction from a supplier-typical parameter prior, anchored at the observed first-cycle SoH. Dotted line: end-of-life threshold (SoH = 0.80). Per-cell in-window SoH RMSE: 0.53, 1.10, 0.97 pp; cross-supplier RMSE std = 0.24 pp.

The supplier-agnostic behaviour appears to come from conditioning on identified electrochemical parameters rather than raw characterisation traces: every supplier enters the operator through the same physical channel. Expanding the anchor pool and per-anchor batched training are the natural next steps.

## References:

1. V. Sulzer, S. G. Marquis, R. Timms, M. Robinson, S. J. Chapman, “Python Battery Mathematical Modelling (PyBaMM)”, *Journal of Open Research Software* **9** (2021), 14.
2. R. T. Q. Chen, Y. Rubanova, J. Bettencourt, D. Duvenaud, “Neural Ordinary Differential Equations”, *Advances in Neural Information Processing Systems* (2018).